MINN2020A Khwezi Mining Report

Reatile Motang (2960250) Simphiwe Nkwanyana (2840655)
Tumelo Moatlhodi (2707368)
Humbulani Maselesele (2989219)

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Abstract

Mining operations generate large volumes of health, safety and equipment data from multiple sources, making it difficult for supervisors and engineers to obtain a consolidated view of operational status. This project presents the **Khwezi Mining Monitor**, a prototype web application designed to centralise worker safety data, safety incidents, equipment condition, and maintenance records into a single interactive platform. The application was developed in **Python 3.10** using **Streamlit** for the interface, **Pandas** for data analysis, **Plotly** for interactive visualisations, and **CSV/JSON** files for storage. It implements role-based access control for five roles, monitors equipment temperature and vibration against predefined thresholds, computes availability using the formula $\text{Availability} = \frac{\text{Operating Time}}{\text{Downtime+Operating Time}} \times 100$, calculates risk using the score $\text{Likelihood} \times \text{Consequence}$, and automatically generates alerts with recommended actions. The result is a functional prototype that demonstrates how programming, data analysis, and visualisation can improve safety management and equipment reliability in a modern mining environment.

1 Introduction

Everyday, critical mining data is scattered across reports, inspections and maintenance records. By the time supersvisors connect the dots, it could be too late. A centralized monitoring application that can collect and analyse relevant mining information can be used detect issues. The Khwezi Mining Monitor brings everything together in one place, it gives real-time insight to identify risk early,monitor incidents, track equipment conditions, identify equipments in need of maintenance and generate alerts when predefined conditions are exceeded.

The increasing availability of digital technologies, sensors and data analytics provides opportunities for mining companies to improve monitoring of health, safety and equipment condition. Rather than relying exclusively on manual inspections and paper-based records, mining personnel can use computer applications to collect, process, visualise and interpret operational data.

This report documents the design, development and evaluation of the Khwezi Mining Monitor, a prototype application developed for the MINN2020A Computer Programming for Mining course. The application transforms mining data into meaningful information that supports timely and informed decision-making by supervisors, engineers and managers.

2 Background Information

A management review at Khwezi Mining identified that current monitoring processes rely on information collected from disparate sources, including safety observations, incident reports, equipment inspections, maintenance records and operator reports. This fragmentation makes it difficult to obtain a consolidated view of the current health and safety status of the operation and the condition of critical mining equipment.

The company therefore required a centralised monitoring application capable of:

- collecting and analysing relevant health, safety and equipment information;
- identifying safety risks and monitoring incidents;
- tracking equipment condition and identifying equipment requiring maintenance;
- generating alerts when predefined conditions are exceeded; and

- providing an interactive dashboard and analytics capability for decision support.

3 Methodology

3.1 System Design and Development Approach

The Khwezi Mining Health, Safety and Equipment Monitoring Application was developed as a Python-based prototype to demonstrate how programming, data analysis, and visualization can support mining health and safety management and equipment reliability. The development process involved identifying the requirements specified in the project brief, generating mining-related datasets, implementing monitoring algorithms, developing an interactive graphical user interface, and integrating the different functions into a centralized application.

The application was developed using Python and and the Streamlit framwork. Streamlit was selected to create and interactive web based interface, while Pandas and Numpy were used for data manipulation and numerical calculations. Plotly was used to produce interactive charts and graphical representations of mining data. The application was designed around seperate functional, maintenance, risk assessment, alerts, analytics and reporting.

3.2 Data Generation and Storage

Since the protoype was deveolped for demonstration purposes, and simulated mining data were generated using python rather than collected from an operating mine. The data generation function created four datasets covering user accounts, worker safety, safety incidents, mining equipment and maintenance records. These datasets stored locally in CSV and JSON formats.

The worker dataset contains 60 simulated worker records, including worker identification, department, job role, shift, personal protective equipment(PPE) compliance, safety training status, fatigue level, safety observations, near misses, previous incidents and risk classification. The incident dataset contains 120 simulated incidents with information about the date, time, location, department, incident type, severity, injury status, cause, corrective action and incident status. The equipment dataset contains 32 simulated machines across eight equipment categories, including haul trucks, loaders, excavators, drills, bulldozers, scraper winches, crushers and conveyors. A separate maintenance dataset contains 40 simulated maintenance records.

They Python random and NumPy random-number generation functions were used to create sample values. Random seeds were set to 42 to support reproducibility when the datasets are generated under the same conditions. CSV files were used to store operational records; JSON was used to store user account information. Pandas was used to load the CSV files into DataFrames for filtering, calculations, grouping, and visualization.

3.3 User Authentication and Access Control

The application uses a login sytem to authenticate users. Passwords are hashed using SHA-256 and stored in a JSON fie. Five user roles are supported, Administrator, Safety Officer, Mining Engineer, Maintenance Engineer and Manager. Each Role has specific permissions to control access to the application's features.

3.4 Worker Safety and Incident Monitoring

The system stores worker information, including PPE compliance, training status, fatigue level and risk categories. it also records mining incidents, their causes, severity and corrective actions. This information helps users identify safety concerns and monitor incident trends.

3.5 Equipment Condition and Maintenance Monitoring

Equipment condition is monitored using temperature and vibration readings. Temperature above 100 degrees Celsius and vibration above 88 mm/s are classified as critical. The system also tracks maintenance, downtime and equipment availability, calculated as:

Availability (percent) = Operanting Hours / (Operating Hours + Downtime)*100

Equipment with availability below 80 percent or downtime above 300 hours is flagged for attention. Sensor readings can be simulated through the application.

3.6 Risk Assessment And Alert Generation

The Application supports the identification of potential safety and equipment risks. Risk levels are used to highlight concerns that require attention, while alerts indicate critical equipment conditions.

3.7 Dashboard, Data Analysis and Reporting

The dashboard presents key information about workers, incidents and equipment. Pandas is used to process and analyse the data, while Ploty creates interactice charts. Users can view summary reports and export selected information in CSV format.

3.8 Interface Design and Implementation

The Application was developed using Python and Streamlit, with custom CSS for its interface. Pandas and NumPy support data processing, while Plotly Displays charts. CSV files store mining records, and a JSON file stores user information. The application can run using the command streamlut run app.py.

4 Results

The completed application runs locally with the command `streamlit run app.py` and is accessed through a web browser at `http://localhost:8501`. The following sections summarise the key outputs.

4.1 Login and Welcome Screen

After authentication, a welcome screen is displayed with the user's name and role, and a single button leading to the dashboard.

4.2 Mining Monitoring Dashboard

The dashboard presents twelve key performance indicators and four interactive charts summarising incidents by department and severity, equipment maintenance status, and average equipment availability by type.

4.3 Condition Monitoring and Alerts

Equipment exceeding the critical temperature or vibration thresholds is displayed with a red alert box matching the required format, including the equipment ID, parameter value and recommended action.

4.4 Interactive Analytics

The analytics page allows the user to select from fifteen business questions, each of which produces a tailored table and chart derived directly from the underlying data.

4.5 Summary of Features Delivered

Requirement	**Delivered**
Access control (RBAC)	Login portal with five roles; two-layer permission checks
Worker health & safety	Filters, unsafe-condition flagging, fatigue/PPE/training alerts
Safety incident database	Add, view, search, filter, count, trend analysis
Equipment database	Central repository with add form and filters
Condition monitoring	Threshold classification with critical-alert boxes
Maintenance monitoring	Availability formula, priority list, low-availability detection
Risk assessment	Likelihood $\times$ Consequence, matrix, classification, warnings
Alerts and notifications	Automated alerts with recommended actions
Monitoring dashboard	12 KPIs, 4 charts, top critical alerts
Interactive analysis	15 Appendix A questions with dynamic charts

5 Conclusions And Recommendations

The Khwezi Mining Monitor demonstrates that a compact, well-structured Python application can consolidate health, safety and equipment data into an interactive monitoring platform aligned with the requirements set out in the project brief. The prototype successfully implements role-based access control, worker and incident monitoring, equipment condition and maintenance analysis, risk scoring, automatic alerts and an interactive dashboard.

Several recommendations follow from the development experience:

- **Migrate to SQLite.** CSV storage is simple and portable but unsuitable for concurrent write access. A lightweight database would support multiple simultaneous users and provide transactional integrity.
- **Add user training** Supervisors need to trust and use the tool. A short training plan can help adopt the app
- **Trend dashboard** Shows incidents and downtime trend over weeks.
- **Add automated notifications.** SMS alerts people who are not looking at the dashboard.
- **Integrate live sensor data.** The current prototype uses synthetic demo data. Connecting to real telemetry from SCADA or IoT platforms would enable continuous monitoring.
- **Add machine-learning anomaly detection.** Time-series analysis of vibration and temperature would detect degradation earlier than static thresholds alone.
- **Improve security.** Replace simple hashed passwords with a proper authentication provider (OAuth, Active Directory) and add audit logging of all user actions.
- **Deploy to a managed cloud environment.** Streamlit Community Cloud, Azure App Service or a containerised deployment would allow multi-user access without local installation.

The prototype meets its stated objective: to demonstrate how programming, data analysis and visualisation can be applied to improve safety management and equipment reliability in a modern mining environment.

6 References

- Numpy(n.d) NumPy documentation. Available at: httpts://numpy.org/doc/stable
- African Review (2026) New reports outline roadmap for South African mining modernisation. African Review of Business and Technology, 15 July. Available at: https://www.africanreview.com
- HSRC (2026) *South Africa's Mining Future Takes Centre Stage at Modernisation Day*. Human Sciences Research Council, 12 June. Available at: https://www.hsrc.ac.za
- PwC South Africa (2026) *Ten Insights into 4IR in South African Mining*. PwC Publications, 20 May. Available at: https://www.pwc.co.za

Appendix: Work Breakdown and Signatures

The following table records the distribution of work across group members. Each member confirms their contribution by signing below.

Task	**Responsible Member**	**Signature**
Data model design and generation	REATILE MOTANG	R.M
Access control and login	SIMPHIWE NKWANYANA	S.N
Worker health and safety module	SIMPHIWE NKWANYANA	S.N
Safety incident database	HUMBULANI MASELESELE	M.P.H
Equipment database and condition monitoring	TUMELO MOATLHODI	T.M
Maintenance monitoring and availability	HUMBULANI MASELESELE	M.P.H
Risk assessment and classification	HUMBULANI MASELESELE	M.P.H
Alerts engine	TUMELO MOATLHODI	T.M
Dashboard and analytics pages	REATILE MOTANG	R.M
Flowcharts and pseudocode	SIMPHIWE NKWANYANA	S.N
Advertising video	REATILE , TUMELO	R.M , T.M
Report writing (Overleaf/LaTeX)	ALL	ALL
GitHub repository management	REATILE MOTANG	R.M
Interview preparation	ALL	ALL

REATILE R.M ..... SIMPHIWE NKWANYANA S.N HUMBULANI MASELESELE M.P.H .....

Name and Signature Name and Signature Name and Signature

TUMELO MOATLHODI T.M . Name and Signature